

CUHK MSC Fintech

FTEC 5910 Industrial Project

Term 1 Presentation

Project Sponsor: Bank of Communications Co.,Ltd. Hong Kong Branch

Topic: Large Language Model Building for Bankwise Processes

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Background

- ▶ Bell was launched in 2018 as an internally developed chatbot by Bank of Communications (BOCOM) R&D team.
- ▶ It uses traditional chatbot technologies and currently provides the following functions:
 - ▶ Daily IT issue
 - ▶ Account unlocking
 - ▶ Contact directory lookup
- ▶ The R&D team plans to integrate Generative AI into Bell to advance the bank's AI capabilities.

Objectives

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- ▶ Goal:
 - ▶ Develop a chatbot system for BOCOM to enhance **document searching** and **Q&A functionalities** of Bell.
- ▶ Technical
 - ▶ Incorporate **Retrieval-Augmented Generation (RAG)** techniques and **Large Language Models (LLMs)** to retrieve information and regulatory requirements efficiently.

Risk and Challenges for Existing Open-source Chatbot System

- ▶ **Language Proficiency Risk:**
 - ▶ LLMs used for those system are primarily trained on English datasets and often underperform on Chinese tasks.
- ▶ **Data Privacy Risk**
 - ▶ They usually used Closed-source or cloud-based LLMs that may expose sensitive internal documents, creating confidentiality concerns
- ▶ **Domain-Specific Knowledge Challenge**
 - ▶ LLMs used for those system may not have sufficient training on banking-specific terminology or regulatory requirements

Hardware for this Project

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Component	Specification	Impact
Accelerator	4x NVIDIA 1080Ti GPUs	Limited VRAM may increase response time for: Multiple interactions with the LLM Large token inputs during queries
GPU VRAM	11,264 MB x 4 =~45 GB	Insufficient CUDA memory for running or fine-tuning large-size LLMs (e.g., Llama 3.2 90B, Qwen 2.5 72B)
Motherboard Memory	900 GB	Adequate for storing multiple LLMs and datasets handling

Hardware Provider: CUHK Fintech Department

Solutions to the Risks and Challenges

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▶ **Chinese Dataset Compatibility**

- ▶ Use LLMs trained on Chinese datasets to ensure high performance in Chinese retrieval tasks.
- ▶ Perform experiments to benchmark their effectiveness

▶ **Data Privacy Assurance**

- ▶ Deploy open-source LLMs on local servers to prevent sensitive data from being exposed to external providers

▶ **Retrieval-Augmented Generation (RAG)**

- ▶ Reduces hallucination by grounding the LLM's outputs with relevant information

▶ **Hardware Optimization**

- ▶ Adapt smaller parameter-sized LLMs that fit within the GPU resources available at CUHK Fintech

Step-by-Step Development Progress

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▶ Term 1:

- ▶ **Experiment 1:** Comparison of Open-Source LLMs to ensure the selection of a model suitable for Chinese RAG tasks.
- ▶ **Experiment 2:** Comparison of RAG Framework to identify the most effective retrieval mechanism for document search.

▶ Term 2:

- ▶ **LLM Fine-Tuning** to adapt the selected model to the domain-specific requirements of Finance and Banking
- ▶ **Prototype Development** to ensure a functional chatbot system for real-world testing in BOCOM's environment

Retriever: Chuxin-Embedding

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- ▶ **Model Size:** 568M parameters
- ▶ **Optimized for Chinese Language:** Specifically designed to handle the nuances of **Chinese text**, ensuring superior retrieval accuracy compared to general-purpose embedding models.
- ▶ Top Performance on Chinese Retrieval Benchmarks:
 - ▶ Ranked #1 on the C-MTEB Chinese Retrieval Leaderboard.
 - ▶ Ranked #1 on the AIR-Bench Chinese Retrieval + Re-ranking Public Leaderboard.
- ▶ **Search Type:** Use similarity-based search to retrieve the most relevant document chunks.
- ▶ **Chunks:** Retrieves the top 3 most relevant chunks from the document corpus for each query, ensuring concise and accurate results.

Dataset

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Topic	Language	Size (kb)	Tokens
Corporate Governance of Locally Registered Authorized Institutions	Chinese	771.7	38912
Climate Risk Management	English	490.1	35840
Guidelines on Combating Money Laundering and Counter-Terrorist Financing	Chinese	1100	84992
Risk-based Regulatory System	Chinese	672.0	30720
Outsourcing	Chinese	173.9	14336
Regulatory Approach to Combating Money Laundering and Terrorist Financing	Chinese	708.9	20480
Risk management of electronic banking	English	337.0	37888
General Principles of Technology Risk Management	Chinese	261.9	29696
Continuing business operations planning	Chinese	551.3	24576
Total		5 mb	317440

Source: HKMA Supervisory Policy Manual

➤ Finance Regulatory Domain

Language :
7 in Chinese, 2 in English

➤ Mainly Chinese with small proportion of English Content

Memory Size: 5 MB

Tokens Count by Retriever:
317,440 tokens

Dataset is selected by project sponsor

A magnifying glass is positioned over a DNA microarray, which displays a grid of colorful spots (red, green, blue, yellow) representing genetic data. The magnifying glass's lens is focused on a specific area, highlighting the intricate details of the microarray. The background is a dark teal gradient.

Experiment 1:

► Comparison of Open-Source LLMs

Description

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- ▶ Objective: Conduct a comprehensive comparison of various open-source LLMs to evaluate their performance in RAG tasks.
- ▶ Test Set: 27 Questions
 - ▶ 3 questions generated from each of the 9 PDFs in the dataset.
- ▶ Methodology
 - ▶ **RAG & No RAG** (Baseline)
- ▶ Evaluation Metrics
 - ▶ **Accuracy** (METEOR) & **Efficiency** (Response Time)
- ▶ Goal: Select the best 3 LLMs on RAG task for future progress

Large Language Models

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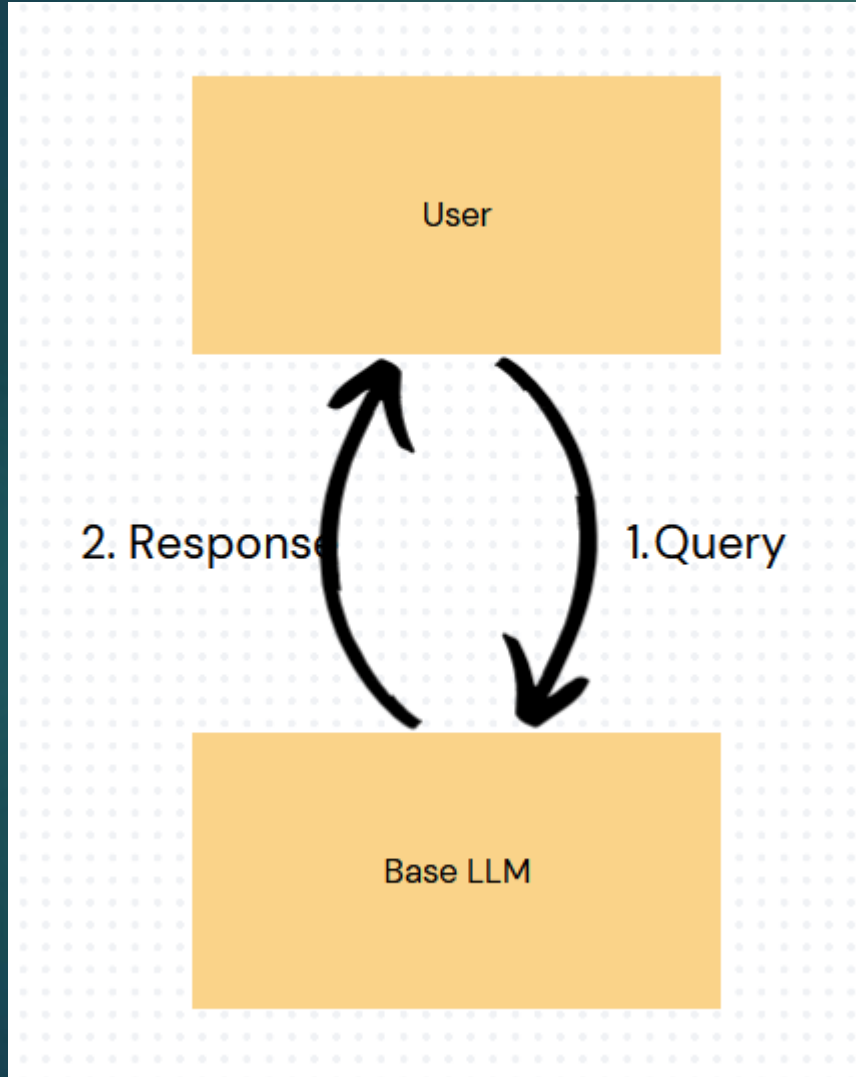
	Llama 3.2	Gemma 2	Qwen 2.5	Yi 1.5
Company:	Meta AI (US)	Google (US)	Alibaba Cloud (China)	01-AI (China)
Released Date	09/2024	06/2024	09/2024	05/2024
Model Size: (billion)	1b/3b/11b	2b/9b	0.5B/1.5B/3B/7B	6b/9b
Maximum Tokens	128,000	8192	3,200/128,000	128,000
Supported Language (Official)	English	English Chinese	English Chinese	English Chinese

For model size: Only limited models can be used in this experiment due to hardware limitation

For Llama 3.2, although this model series officially does not support Chinese. However, this model is trained by a large collection of languages which includes Chinese.

Methodology: No RAG (Baseline)

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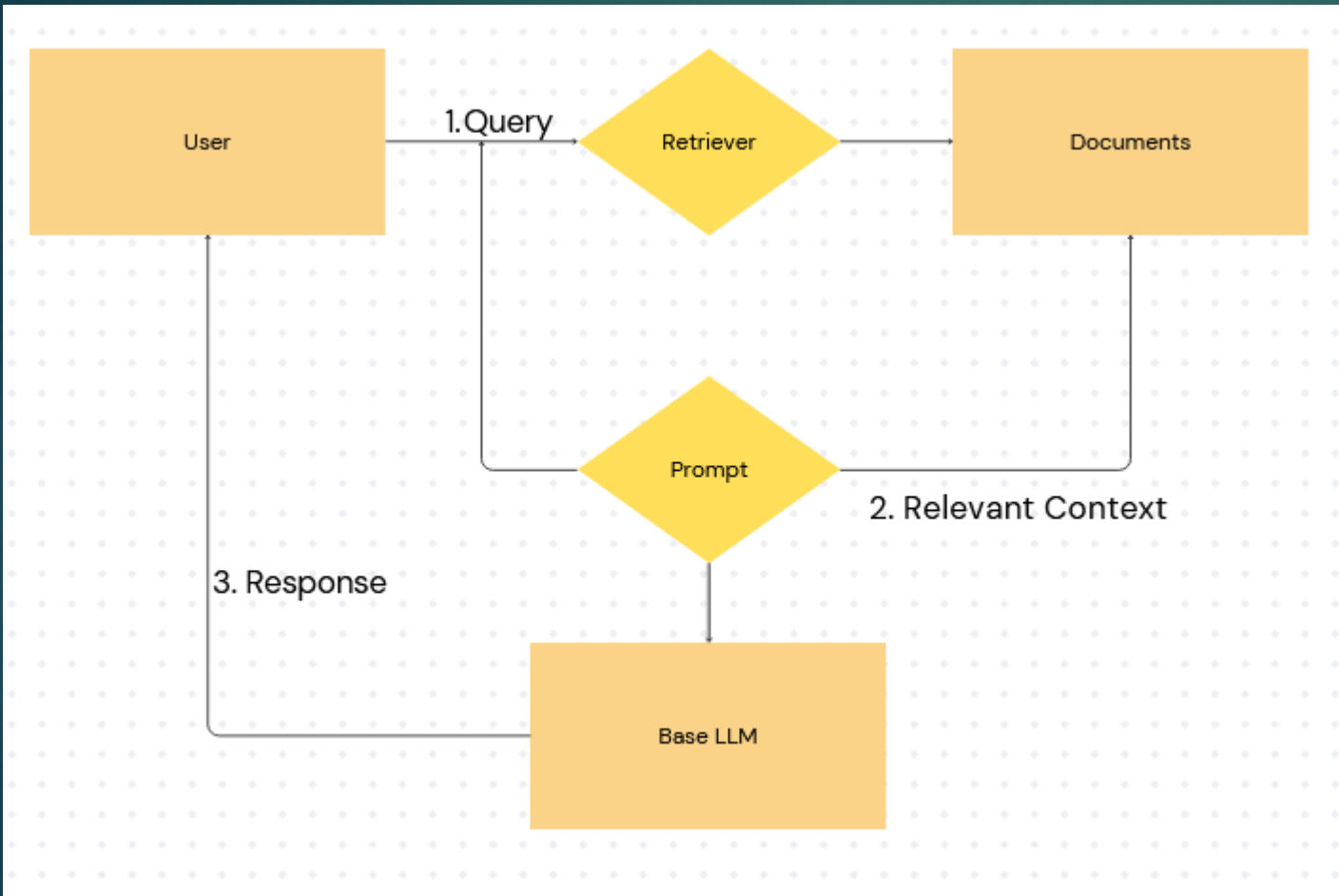
Summary: Generate responses without incorporating external document retrieval.

Role:

- ▶ Serves as the baseline to compare performance with the RAG approach.
- ▶ Help to evaluate how much RAG method improves the model's accuracy on finance regulatory task

Methodology: RAG

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Summary: Generate responses based on retrieved context.

Role:

- Evaluates how much RAG method improves model performance for accuracy and efficiency

Prompt Template

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No RAG (Baseline)	RAG
“””” 使用不超過200字繁體中文(連標點)來回答問題。 Question: {question} Answer:	“””” 使用不超過200字繁體中文(連標點)回答所給的上下文來回答問題。 上下文: {Relevant context} Question: {question} Answer:

For RAG prompt, each model shares the same relevant document for each question

Evaluation Metrics

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Accuracy: Metric for Evaluation of Translation with Explicit Ordering (METEOR)

1. → METEOR Score (Model Response ↔ Target Response) ↵

$$P = \frac{m}{w_t}, R = \frac{m}{w_r}, F_{mean} = \frac{10PR}{R + 9P} \downarrow$$

$$Penalty = 0.5 \left(\frac{C}{U_m} \right)^3, METEOR = F_{mean} (1 - Penalty) \leftarrow$$

where: ↵

m : Number of unigrams in the model response also found in target response ↵

w_t : Number of unigrams in the model response ↵

w_r : Number of unigrams in target response ↵

C : Number of chunks in the model response ↵

U_m : Unigrams in the model response ↵

Equation 1: Formula of METEOR Score ↵

Efficiency: Response Time

Result

Input tokens for
all LLMs are the
same

Metrics↵	Accuracy↵		Response Time↵		Output Tokens↵	
Method↵	No-RAG↵	RAG↵	No-RAG↵	RAG↵	No-RAG↵	RAG↵
Llama-3.2-1B↵	0.0672↵	0.2334↵	1.81s↵	6.41s↵	55↵	80↵
Llama-3.2-1B-Instruct↵	0.1270↵	0.3251↵	2.19s↵	6.07s↵	96↵	97↵
Llama-3.2-3B↵	0.0699↵	0.2456↵	6.85s↵	12.74s↵	86↵	142↵
Llama-3.2-3B-Instruct↵	0.1332↵	0.4693↵	10.04s↵	13.81s↵	187↵	182↵
Llama-3.2-11B↵	0.1616↵	0.4362↵	11.67s↵	24.89s↵	200↵	200↵
Llama-3.2-11B-Instruct↵	0.1949↵	0.4984↵	10.52s↵	17.67s↵	173↵	111↵
Gemma-2-2B↵	0.0878↵	0.2080↵	13.22s↵	15.07s↵	110↵	195↵
Gemma-2-2B-it↵	0.1393↵	0.4081↵	6.04s↵	7.56s↵	131↵	126↵
Gemma-2-9B↵	0.1233↵	0.2363↵	21.96s↵	30.00s↵	138↵	180↵
Gemma-2-9B-it↵	0.1816↵	0.5045↵	11.89s↵	16.37s↵	99↵	70↵
Qwen-2.5-1.5B↵	0.1401↵	0.3557↵	5.74s↵	6.44s↵	91↵	94↵
Qwen-2.5-1.5B-Instruct↵	0.1721↵	0.3588↵	2.19s↵	6.44s↵	96↵	101↵
Qwen-2.5-3B↵	0.2062↵	0.4161↵	3.48s↵	10.56s↵	69↵	66↵
Qwen-2.5-3B-Instruct↵	0.1849↵	0.4197↵	1.81s↵	9.30s↵	55↵	90↵
Qwen-2.5-7B↵	0.1816↵	0.2164↵	16.03s↵	24.93s↵	154↵	193↵
Qwen-2.5-7B-Instruct↵	0.1342↵	0.3631↵	10.48s↵	21.19s↵	109↵	126↵
Yi-1.5-6B↵	0.1087↵	0.1906↵	16.30s↵	28.56s↵	166↵	175↵
Yi-1.5-6B-Chat↵	0.1519↵	0.3680↵	16.37s↵	22.07s↵	180↵	196↵
Yi-1.5-9B↵	0.1752↵	0.2842↵	7.70s↵	19.26s↵	101↵	100↵
Yi-1.5-9B-Chat↵	0.1806↵	0.4757↵	7.41s↵	20.41s↵	99↵	99↵

Question

電傳轉帳是甚麼？

Retrieved context: Top 3 chunks

電傳轉帳是由一間機構(匯款機構)代表某人(匯款人)藉電子方式進行的交易，目的是將某筆金錢轉往某間機構(收款機構) (該機構可以是匯款機構或另一機構)以提供予該人或另一人(收款人)，而無論是否有一間或多於一間 其他機構(中介機構)參與完成有關金錢轉帳。

認可機構 應就其在電傳轉帳中扮演的角色，遵從本章所載的相關規定

如認可機構是電傳轉帳的匯款人或收款人，即並非以匯款機構、中介機構或收款機構的身分行事，則無需就該交易遵守附表2 第12 條的規定或本章的指引。

Model	Gemma 2 9B it		Llama 3.2 11B Instruct		Yi 1.5 9B Chat	
Question	電傳轉帳是甚麼？					
Target Response	電傳轉帳是由一間機構代表某人藉電子方式進行的交易，目的是將某筆金錢轉往某間機構以提供予該人或另一人，而無論是否有一間或多於一間其他機構參與完成有關金錢轉帳。					
Method	No RAG	RAG	No RAG	RAG	No RAG	RAG
LLM Response	電傳轉帳，簡稱電匯，是一種通過銀行電訊網路進行資金轉移的方式。它比傳統的支票或匯票更快更有效率，因為資金幾乎是即時轉帳的。你需要提供收款人的銀行帳戶資訊，例如帳號、銀行名稱、SWIFT代碼（國際銀行識別碼）等等，才能完成電匯。電匯通常會收取一定的費用，費用會根據轉帳金額和銀行而有所不同。總之，電匯是一種方便快捷的國際或國內資金轉帳方式。	電傳轉帳是一種電子交易方式。匯款機構代表匯款人，將款項透過電子方式轉到收款機構（可能是匯款機構本身或其他機構），最終交付給收款人。過程中可能涉及其他中介機構參與。認可機構若僅作為匯款人或收款人，而非扮演匯款機構、中介機構或收款機構的角色，則無需遵守相關規定。簡單來說，電傳轉帳就是一種電子化的資金轉移方式。	電傳轉帳是一種電子支付方式,用於轉移電子金錢。它允許用戶跨銀行或支付系統進行即時電匯交易,簡化了傳統匯款流程。	電傳轉帳是由匯款機構代表匯款人以電子方式進行的交易，將資金從一個國家的銀行帳戶轉至另一個國家的銀行帳戶，通常透過SWIFT系統完成，具備快速和安全的特點，但可能涉及手續費。如果認可機構是以匯款人或收款人的身分進行電傳轉帳，而非作為匯款機構、中介機構或收款機構，則無需遵守附表2第12條的規定或相關指引。整個過程可包括一間或多間中介機構協助完成資金轉移。	電傳轉帳是一種透過電子網路將資金從一個銀行帳戶轉移到另一個銀行帳戶的付款方式。它是一種快速、安全且方便的轉帳方式，通常比傳統的支票或現金轉帳更快。電傳轉帳通常需要提供轉帳人的銀行帳戶資訊，例如帳戶號碼和銀行代碼，以及收款人的銀行帳戶資訊。	電傳轉帳是指透過電子方式，由一間機構代表某人將金錢轉往另一機構，以提供給特定收款人。
METEOR	0.1518	0.2902	0.1097	0.3145	0.1706	0.2625

Discussion (LLMs Comparison)

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- ▶ Top 3 LLMs on RAG Accuracy:
 - Gemma 2 9B it
 - Llama 3.2 11B Instruct
 - Yi 1.5 9B Chat
- ▶ Instructed-tuned models generally have better accuracy in both no-rag and rag tasks than their base model
- ▶ 200 maximum output tokens maybe proper
 - ▶ Since top three LLMs on RAG accuracy do not have the largest output tokens

Discussion (No RAG vs RAG)

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▶ **No RAG Response (Baseline):**

- ▶ Generally have lower accuracy and higher efficiency than RAG
- ▶ Limited Knowledge
 - ▶ Relies only on the LLM's pre-trained knowledge
 - ▶ Example: Struggles to provide precise details about "telegraphic transfer" without external context.

▶ **RAG Response:**

- ▶ Generally have higher accuracy and lower efficiency than RAG
- ▶ Incorporates Relevant Context
 - ▶ Retrieves domain-specific documents to ground the LLM's responses
 - ▶ Example: RAG responses included precise explanations about telegraphic transfers by leveraging retrieved content.

Limitation of Basic RAG Method

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- ▶ **Faithfulness:** The response may omit key details from the retrieved content
 - ▶ E.g. Response from Llama (RAG) the potential involvement of intermediary institutions ("中介機構"). It only partially explains the roles of remitting and receiving institutions.
 - ▶ Missing key elements lowers recall, as the response fails to include all relevant details present in the target response Limited Knowledge
- ▶ **Answer Relevant:** The response maybe verbose, over-elaborating on intermediary institutions and roles. Incorporates Relevant Context.
 - ▶ E.g. "中介機構也可能參與" in Llama (RAG) (intermediary institutions may also participate) is accurate but unnecessarily elaborated.
 - ▶ Verbose responses include extraneous content that reduces precision and alignment with the target response's brevity.



Experiment 2:

► Comparison of Different RAG Methods

Description

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- ▶ Objective: Comparison of different RAG frameworks to enhance the overall performance of the chatbot system
- ▶ Test Set: 27 Questions (Same as experiment 1)
- ▶ Methodology
 - ▶ **RAG** alternatives
- ▶ Evaluation Metrics
 - ▶ **Accuracy, Efficiency** (Same as experiment 1)
 - ▶ **Generation Qualities** (Faithfulness & Answer Relevant)
- ▶ Goal: Select the best RAG framework for this chatbot system

Evaluation Metrics

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Generation Quality (Faithfulness) Formula:

2. → Faithfulness (Retrieved Context ↔ LLM Response) ↵

$$Faithfulness = \frac{C}{T} \downarrow$$

where: ↵

- C is Number of claims in model's response that are consistent with the retrieved context, ↵
- T is Total number of claims in model response ↵

Equation 2: Formula of Faithfulness Score ↵

- Use LLM as Judge (GPT 3.5)
- Range from 0 to 1
 - 1 indicates that all claims in LLM's response can be inferred from the given context
 - Lower value represent how inconsistent the model's response is from the given context

Prompt to use LLM as Judge

Instruction	Your task is to judge the faithfulness of a series of statements based on a given context. For each statement you must return verdict as 1 if the statement can be directly inferred based on the context or 0 if the statement can not be directly inferred based on the context.
Example Input	Context: "....." Statement: [".....", ".....", ".....", "....."]
Example Output	Statement: "....." Reason: "....." Verdict: (0/1) (Repeat many times)
My Input	Context: "....." Statement: [".....", ".....", ".....", "....."]

Evaluation Metrics

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Generation Quality (Answer Relevant) Formula:

3. → Answer Relevance (User Query ↔ LLM Response) ←

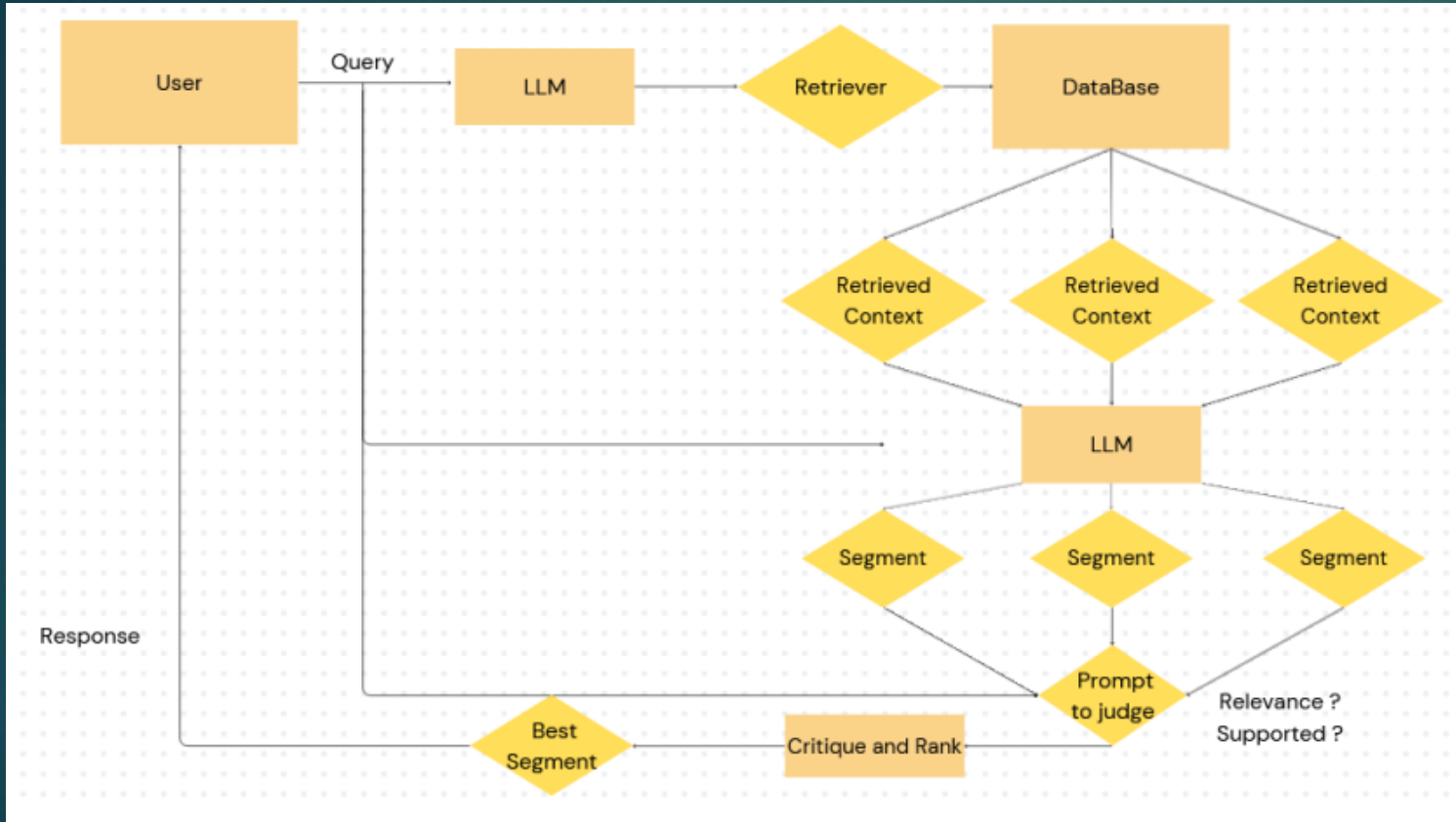
$$\text{Answer Relevance} = \frac{1}{n} \sum_{i=1}^n \text{sim}(q, q_i) \leftarrow$$

Equation 3: Formula of Answer Relevance Score ←

- Use Cosine Similarity to calculate the similarity score
- Range from 0 to 1
 - 1 indicates that all claims in LLM's response can be inferred from the given context
 - Lower value represent how inconsistent the model's response is from the given context

Methodology: RAG

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Summary:

- Generate responses based on each retrieved context
- Iterative self-evaluation based on whether the response is supported by retrieved context and relevant to user's query
- The best-ranked segment is used as the final response

Methodology

Self-Reflective RAG

Prompt to use LLM to judge whether their responses are supported by the retrieved context or relevant to the answer

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Prompt to judge	
Instruction:	You are tasked with evaluating the quality of a response segment based on its relevance and supportiveness with respect to the user query and retrieved document. ISUSE: Assess whether the retrieved document was used in generating the response. This means the response relied on the document to provide its content. Answer with "Yes" or "No". ISSUP: Assess whether the response segment is supported by the retrieved document. A response is supported if it directly uses information from the document or aligns with its content. Answer with "yes" or "no".
Example	User Query: What is the capital of France? Retrieved Context: Paris is the capital city of France and one of its major centers of commerce and culture. Generated Response Segment: The capital of France is Paris. ISSUP: Yes ISUSE: Yes
Input	User Query: {query} Retrieved Context: {Relevant Context} Generated Response Segment: {response} ISSUP: . ISUSE: . *3

Result

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Self-Reflective RAG↵				
Models↵	Accuracy↵	Response Time↵	Faithfulness↵	Answer Relevant↵
Gemma-2-9B-it↵	0.5401↵	80.56s↵	0.8935↵	0.8741↵
LLaMA-3.2-11B-Instruct↵	0.5213↵	78.13s↵	0.8813↵	0.8612↵
Yi-1.5-9B-Chat↵	0.5331↵	75.26s↵	0.8916↵	0.8667↵
↵				
Compared to Baseline with RAG↵				
Models↵	Accuracy↵	Running Time↵	Faithfulness↵	Answer Relevant↵
Gemma-2-9B-it↵	0.5045↵	16.37s↵	0.9135↵	0.7968↵
LLaMA-3.2-11B-Instruct↵	0.4984↵	17.67s↵	0.9068↵	0.8059↵
Yi-1.5-9B-Chat↵	0.4757↵	20.41s↵	0.8864↵	0.8378↵
↵				
Compared to Baseline without RAG↵				
Models↵	Accuracy↵	Running Time↵	Faithfulness↵	Answer Relevant↵
Gemma-2-9B-it↵	0.1816↵	11.89s↵	/↵	0.8376↵
LLaMA-3.2-11B-Instruct↵	0.1949↵	10.52s↵	/↵	0.8302↵
Yi-1.5-9B-Chat↵	0.1806↵	7.41s↵	/↵	0.8203↵

Example

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Model	Gemma 2 9B it		Llama 3.2 11B Instruct		Yi 1.5 9B Chat	
Question	電傳轉帳是甚麼？					
Target Response	電傳轉帳是由一間機構代表某人藉電子方式進行的交易，目的是將某筆金錢轉往某間機構以提供予該人或另一人，而無論是否有一間或多於一間其他機構參與完成有關金錢轉帳。					
Method	RAG	Self-RAG	RAG	Self-RAG	RAG	Self-RAG
LLM Response	電傳轉帳，簡稱電匯，是一種通過銀行電訊網路進行資金轉移的方式。它比傳統的支票或匯票更快更有效率，因為資金幾乎是即時轉帳的。你需要提供收款人的銀行帳戶資訊，例如帳號、銀行名稱、SWIFT代碼（國際銀行識別碼）等等，才能完成電匯。電匯通常會收取一定的費用，費用會根據轉帳金額和銀行而有所不同。總之，電匯是一種方便快捷的國際或國內資金轉帳方式。	電傳轉帳，簡稱電匯，是一種透過電子方式進行的資金轉移。匯款人委託匯款機構，將款項轉給收款機構，再由收款機構交付給收款人。過程中，可能會有其他中介機構參與，但最終目的是將資金從匯款人轉到收款人手中。它是一種快速且便捷的國際或國內資金轉帳方式	電傳轉帳是由匯款機構代表匯款人以電子方式進行的交易，將資金從一個國家的銀行帳戶轉至另一個國家的銀行帳戶，通常透過SWIFT系統完成，具備快速和安全的特點，但可能涉及手續費。如果認可機構是以匯款人或收款人的身分進行電傳轉帳，而非作為匯款機構、中介機構或收款機構，則無需遵守附表2第12條的規定或相關指引。整個過程可包括一間或多間中介機構協助完成資金轉移。	電傳轉帳是一種交易方式, 由一間機構(匯款機構)代表某人(匯款人)藉電子方式進行的交易, 目的是將某筆金錢轉往另一間機構(收款機構)以提供予該人或另一人(收款人), 而無論是否有一間或多於一間其他機構(中介機構)參與完成有關金錢轉帳。	電傳轉帳是指透過電子方式，由一間機構代表某人將金錢轉往另一機構，以提供給特定收款人。	電傳轉帳是一種透過電子方式，由匯款機構代表匯款人將金錢轉至收款機構，提供給收款人或另一人，無論是否透過中介機構完成的交易。
METEOR	0.1518	0.3094	0.3145	0.8386	0.2625	0.3703
Faithfulness	0.5000	0.6667	1	1	1	1
Answer Relevant	0.6400	0.6634	0.6418	0.6958	0.7584	0.7090

Discussion (RAG vs Self-Reflective RAG)

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- ▶ **Generation Quality (Faithfulness):**
 - ▶ Self-reflective RAG is not likely to improve the faithfulness
 - ▶ 2 models have lower faithfulness score when using self-reflective RAG
- ▶ **Generation Quality (Answer Relevance):**
 - ▶ Consistently better with Self-RAG
 - ▶ All 3 models have higher relevant score when using self-reflective RAG
- ▶ **Accuracy (METEOR Score):**
 - ▶ Self-reflective RAG ensures higher accuracy
 - ▶ All 3 models have higher accuracy when using self-reflective RAG

Discussion (Self-Reflective RAG)

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- ▶ **Advantage: Improved Answer Relevant**

- ▶ E.g. Gemma response using basic RAG included unnecessary elaboration (unrelated details about SWIFT codes) and misses the main context (transaction intent)
- ▶ Self-Reflective RAG excludes unnecessary elaboration and focused on the transaction intent

- ▶ **Disadvantage: Increased Computational Cost**

- ▶ Requires iterative critique and ranking, leading to significantly higher processing time
- ▶ E.g., Gemma 2: 80.56s in Self-RAG vs. 16.37s in RAG

Conclusion

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After the above experiments, here is some key results

- ▶ Selected top 3 LLMs for this project
 - Gemma 2 9B it
 - Llama 3.2 11B Instruct
 - Yi 1.5 9B Chat
- ▶ RAG method has better accuracy than No RAG method in this project
- ▶ Self-Reflective RAG has better accuracy and generation quality than RAG, but it costs longer response time

Challenge for this project

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▶ **Long Response Time for Multiple Interactions**

- ▶ the acceptable response time of the project is lower than 10 seconds
- ▶ the latency time using self-reflective RAG usually takes more than one minute for each response

▶ **Hardware Limitations**

- ▶ Response time of Gemma 2 on provided GPU server (Nvidia 1080 Ti) using basic RAG method: 16.37s per response
- ▶ Response time of Gemma 2 on cloud server (POE) using basic RAG method: 1s per response

Further Prospect on Term 2

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- ▶ **Research on other advanced RAG framework at Experiment 2**
 - ▶ To improve the accuracy and generation quality, while addressing the tradeoff between accuracy and computational efficiency
- ▶ **Fine-tuning the LLMs to enhance its RAG capabilities**
 - ▶ Create 200 questions and answers pairs
 - ▶ To improve the LLM's ability to generate accurate, factual and relevant responses based on retrieved context
- ▶ **Building the prototypes**
 - ▶ To showcase the chatbot system's functionality and capabilities at the end of the project

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